Design a VPN (MPLS) Network

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Presentation Agenda

Part 1: Context & Theory

- Introduction
- Motivation, Objectives & Theory

Part 2: Implementation

- System Design
- Topology, IP Plan & Configuration

Part 3: Outcomes

- Results
- Verification & Connectivity Tests
- Conclusion
- Summary & Future Works

Introduction & Project Abstract

The Challenge

- Modern enterprises require secure, reliable, and scalable communication. Traditional VPNs often face limitations in performance and management complexity as networks grow.

The Solution

- We designed an enterprise MPLS VPN topology using PNETLab. Key protocols (LDP, BGP, OSPF) were configured to establish label switching. VRF technology was implemented for strict traffic separation.

Project Abstract

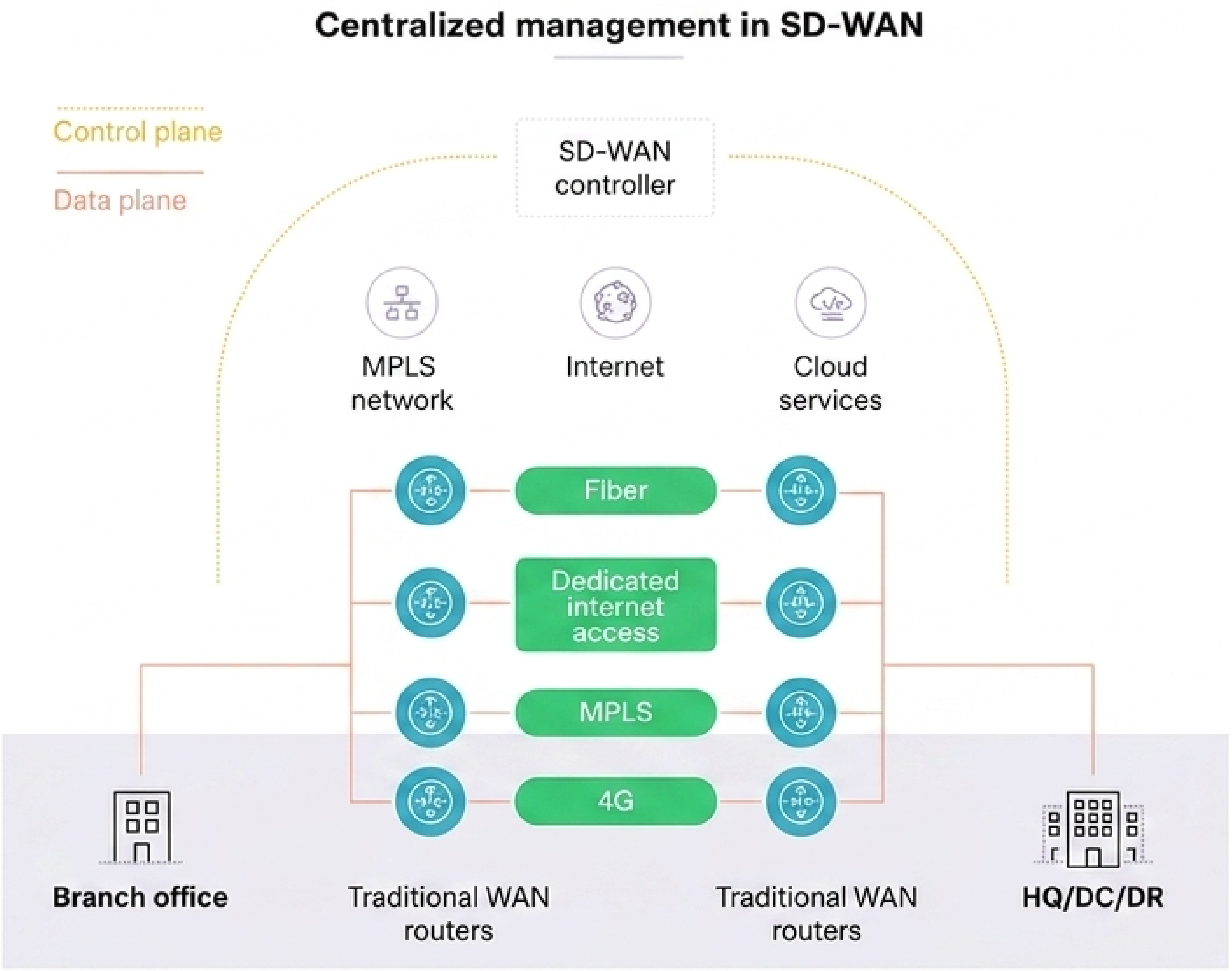
Connecting remote sites efficiently while mitigating the scalability issues of traditional tunneling.

Why MPLS VPN?

- **Scalability:** Avoids the mesh of tunnels in traditional VPNs.
- **Performance:** Combines IP routing scalability with L2 switching speed.
- **Traffic Engineering:** Better management of diverse traffic types.
- **Isolation:** Strict separation via VRF.

Motivation

To build a robust network architecture capable of handling diverse traffic types with varying Quality of Service requirements.



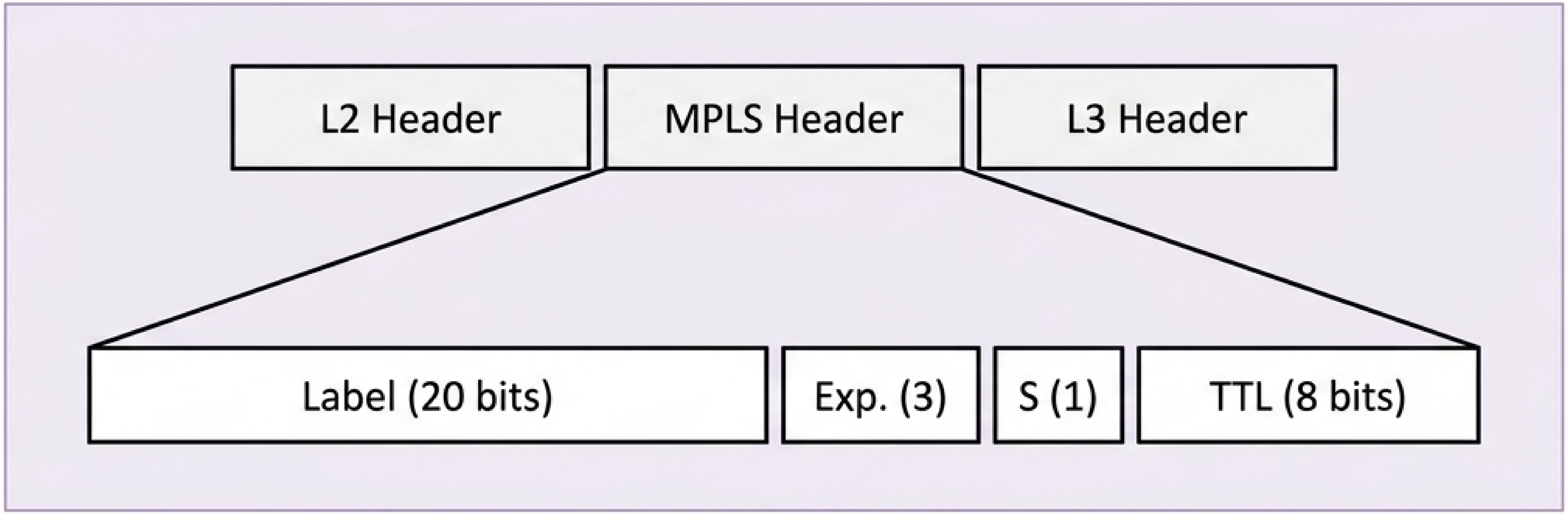
NotebookLM

Project Objectives

Design & Simulation Create a comprehensive enterprise topology using PNETLab and Cisco IOS.	**Protocol Implementation** Configure the core MPLS backbone using LDP and OSPF.
VPN Configuration Implement Layer 3 VPN (L3VPN) services using MP-BGP and VRF.	**Verification** Validate connectivity, traffic separation, and forwarding paths.

MPLS Technology Overview

Multiprotocol Label Switching (MPLS) speeds up traffic flows by using short path labels instead of long IP lookups. Core routers (LSRs) forward data based solely on these labels, allowing for protocol-agnostic transport.



Standard MPLS Header (Label, Exp, S, TTL)

Label Distribution Protocol (LDP)

1

Discovery

Hello messages sent to multicast 224.0.0.2 to find neighbors.

2

Session

Establishes a reliable TCP session on port 646.

3

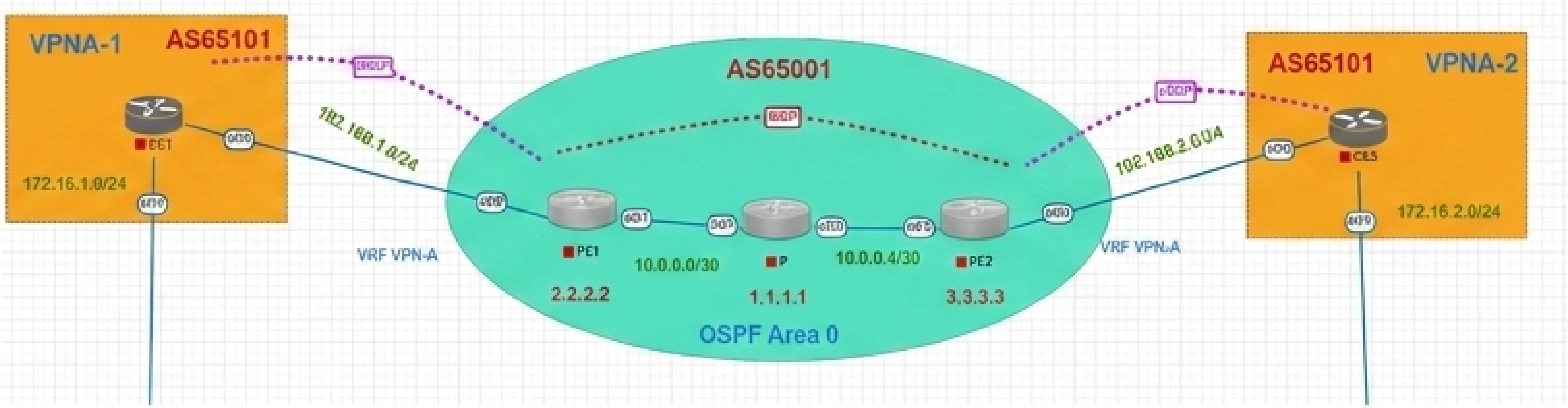
Exchange

Routers exchange label mappings for known networks.

L3VPN Architecture

Key Components:

- **CE (Customer Edge):** Customer router, unaware of MPLS.
- **PE (Provider Edge):** Critical device. Connects to CE, maintains VRF tables, runs MP-BGP.
- **P (Provider):** Core router. Only performs high-speed label switching.



L3VPN Network Architecture Diagram

VRF & MP-BGP

VRF (Virtual Routing & Forwarding)

Allows multiple independent routing tables on a single router.

Ensures "Customer A" traffic never leaks to "Customer B" traffic never leaks to "Customer B", even if they use the same IP addresses.

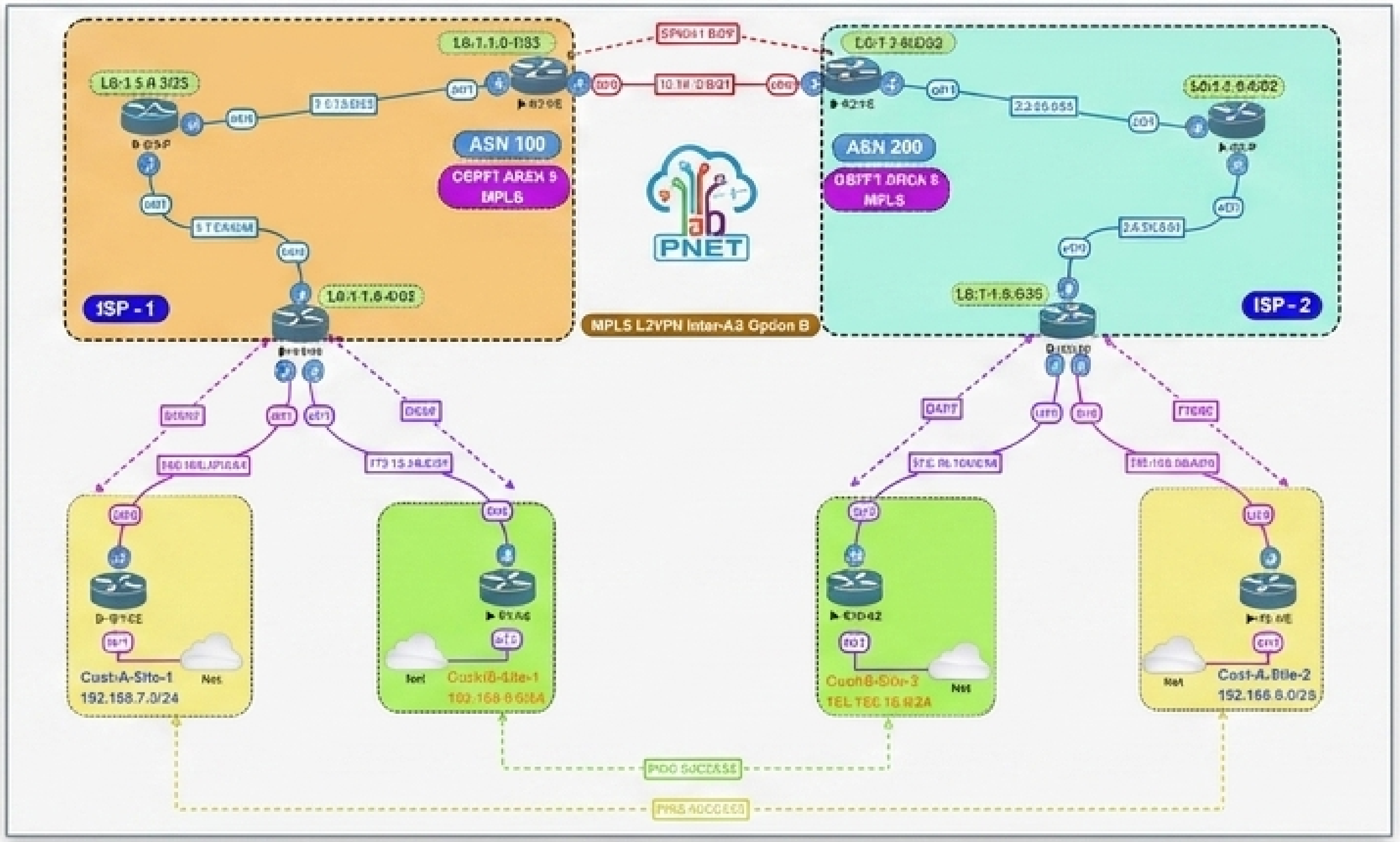
MP-BGP (Multiprotocol BGP)

Standard BGP only carries IPv4.

MP-BGP carries VPNv4 routes (IPv4 + Route Distinguisher) to keep prefixes unique across the backbone.

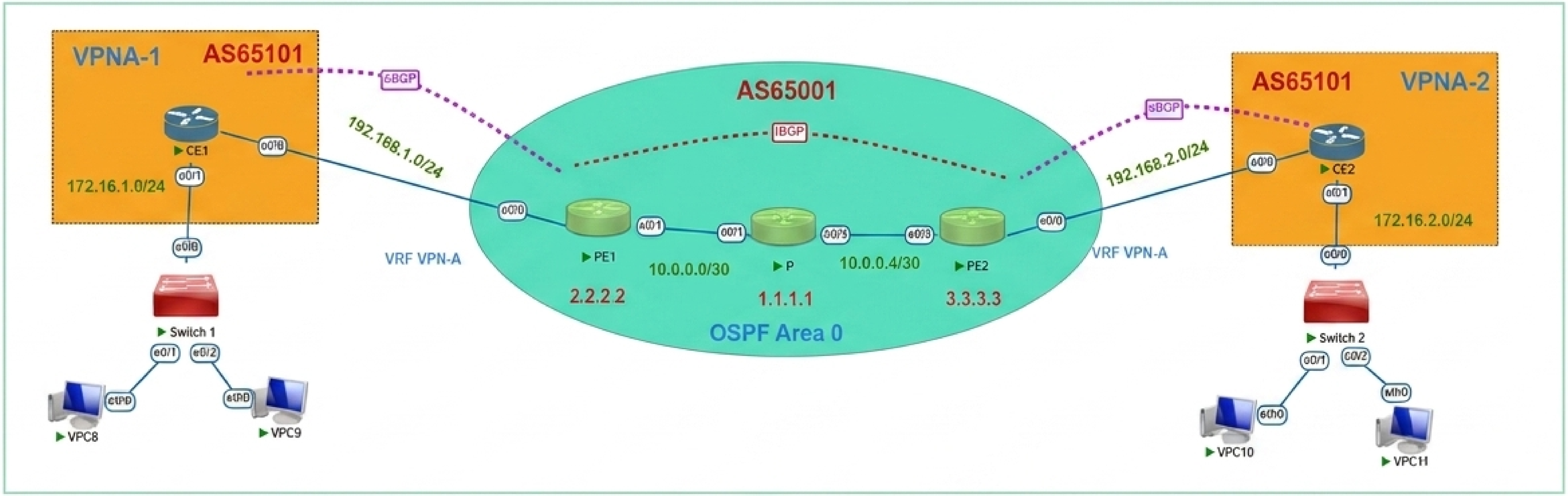
Simulation: PNETLab

- Runs actual Cisco IOSv.
- Visual workspace for topology design.
- Wireshark integration for packet analysis.



PNETLab Workspace

Topology Design: Scenario Overview



Full Network Topology Design

Service Provider (AS 65001):

- 1 P Router (Core)
- 2 PE Routers (Edge)

Customer (AS 65101):

- Site 1 (CE1)
- Site 2 (CE2)

Goal:

- Connect Site 1 and Site 2 securely.

IP Addressing Scheme

Device	Interface	IP Address/Subnet Mask	Description
P Router	Loopback0	1.1.1.1/32	MPLS Router ID
PE1	Loopback0	2.2.2.2/32	BGP Peer ID
PE1	E0/1	10.0.0.1/30	Link to CE1 (VRF)
PE2	Loopback0	3.3.3.3/32	BGP Peer ID
CE1	E0/1	172.16.1.1/24	LAN Gateway Site 1

MPLS Backbone: OSPF

- We use OSPF Area 0 as the IGP to provide full IP reachability between Loopbacks and physical links in the core.
- This is the foundation before enabling MPLS.

R1 Verifying MPLS enabled interfaces on P router.

```
! Configuration on PE1 (Core facing interface)
PE1(config)# interface e0/1
PE1(config-if)# description Link-to-P
PE1(config-if)# ip address 10.0.0.1 255.255.255.252
PE1(config-if)# mpls ip
```

Core interface configuration on PE1

L3VPN: VRF Definition

Route Distinguishers

- **RD (65001:1):** Identifies the VPN locally.
- **RT (65001:1):** Controls Import/Export of routes.

Traffic arriving on E0/0 is now isolated into table 'VPNA'.

```
! Configuration on PE1
PE1(config)# ip vrf VPNA
PE1(config-vrf)# rd 65001:1
PE1(config-vrf)# route-target export 65001:1
PE1(config-vrf)# route-target import 65001:1

PE1(config)# interface e0/0
PE1(config-if)# description Link-to-CE1
PE1(config-if)# ip vrf forwarding VPNA
PE1(config-if)# ip address 192.168.1.2 255.255.255.0
```

VRF Configuration Commands

MP-BGP Configuration

Peering PE1 & PE2

We establish an iBGP session between Loopbacks.

The `address-family vpnv4` and `send-community extended` commands are crucial for carrying the VPN labels and attributes across the core.

```
! Configuration on PE1
PE1(config)# router bgp 65001
PE1(config-router)# bgp router-id 2.2.2.2
PE1(config-router)# no bgp default ipv4-unicast
PE1(config-router)# neighbor 3.3.3.3 remote-as 65001
PE1(config-router)# neighbor 3.3.3.3 update-source Loopback0
PE1(config-router)# address-family vpnv4
PE1(config-router-af)# neighbor 3.3.3.3 activate
PE1(config-router-af)# neighbor 3.3.3.3 send-community extended
PE1(config-router-af)# exit-address-family
```

MP-BGP Configuration on PE1

MP-BGP Session Establishment between PE Routers.

PE-CE Routing Configuration (OSPF/Static)

The Customer Edge (CE) router peers with the Provider via standard eBGP and advertises its local LAN prefix.

Provider & Customer configuration

Note: `as-override` is crucial here to prevent loop detection issues since both customer sites use AS 65101.

Verification: Core Status

LDP Neighbors

The status “Operational” on the P router confirms that label bindings are being exchanged with PE1 and PE2.

Verification: Core Status

The presence of valid LDP neighbors ensures the core is ready for label switching.

```
P>show mpls ldp neighbor
   Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 1.1.1.1:0
      TCP connection: 3.3.3.3.13317 - 1.1.1.1.646
      State: Oper; Msgs sent/rcvd: 65/65; Downstream
      Up time: 00:50:39
      LDP discovery sources:
        Ethernet0/3, Src IP addr: 10.0.0.6
      Addresses bound to peer LDP Ident:
        10.0.0.6        3.3.3.3
  Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 1.1.1.1:0
      TCP connection: 2.2.2.2.21604 - 1.1.1.1.646
      State: Oper; Msgs sent/rcvd: 65/65; Downstream
      Up time: 00:50:27
      LDP discovery sources:
      Ethernet0/1, Src IP addr: 10.0.0.1
      Addresses bound to peer LDP Ident:
        10.0.0.1        2.2.2.2
```

LDP Neighbor Status Output

Verification: VRF Routing

Isolated Tables

The command `show ip route vrf VPNA` confirms that PE1 has learned the remote LAN prefix (172.16.2.0/24) via BGP.

This route exists only inside the VRF, not the global table.

Telnet 192.168.17.130

```
PE1>show ip route vrf VPN-A

Routing Table: VPN-A
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      172.16.0.0/24 is subnetted, 2 subnets
B        172.16.1.0 [20/0] via 192.168.1.1, 01:10:39
B        172.16.2.0 [200/0] via 3.3.3.3, 01:10:39
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, Ethernet0/0
L        192.168.1.2/32 is directly connected, Ethernet0/0
```

VRF Routing Table on PE1

End-to-End Connectivity & Path Analysis

End-to-End Connectivity

- Ping Test
- Source: VPC 8 (Site 1)
- Dest: VPC 10 (Site 2)

The successful ping confirms full reachability between the two remote branches through the MPLS cloud.

Path Analysis: Traceroute

- Label Switching Confirmed

  Hop 3 (P Router): The presence of an MPLS Label in the output proves that the core is switching packets based on labels, not IP addresses.

Project Conclusion

Successful Implementation

We designed and simulated a fully functional MPLS L3VPN.

Robust Backbone

The core network effectively uses LDP and OSPF for fast transport.

Security

VRF and MP-BGP successfully isolated customer traffic.

Validation

Tests confirmed end-to-end connectivity and correct label switching behavior.

Summary & Future Works

QoS Integration

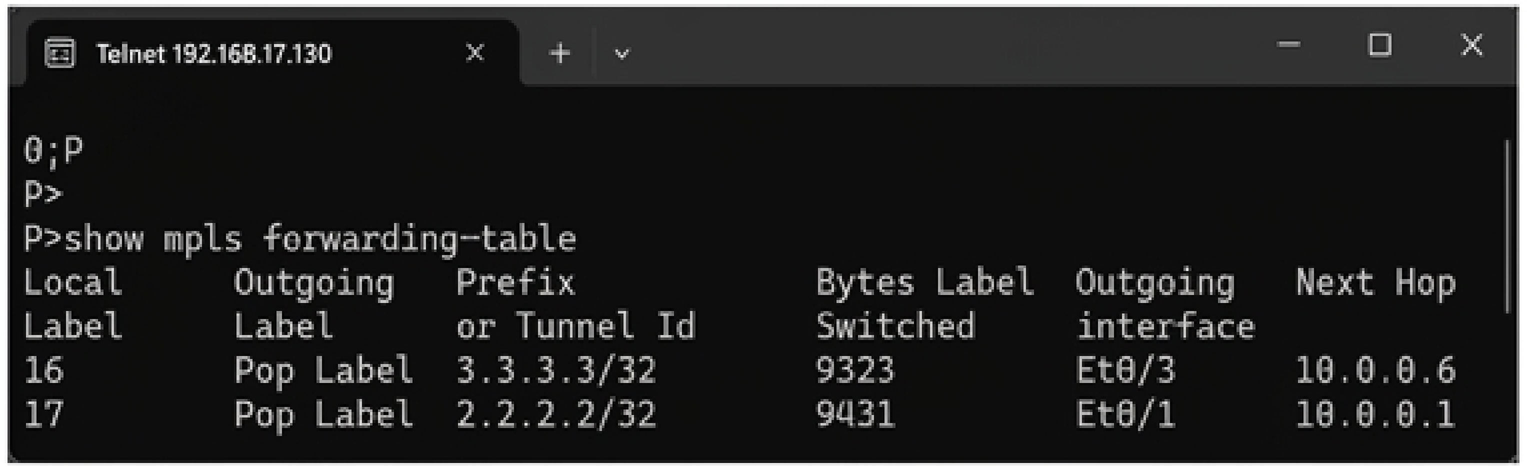
Implement traffic classification to prioritize VoIP and Video.

SD-WAN Hybrid

Integrate MPLS with SD-WAN overlays for a modern hybrid WAN.

IPv6 (6VPE)

Extend the design to support IPv6 VPNs for future compatibility.



MPLS Forwarding Table verifying future-ready label switching